

A Simple Model of Abatement Costs

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The note deals with two of the most important concepts in climate economics: The economic damages related to climate change and the cost of abating emissions to mitigate said damages. In this introductory note, we present the concepts with a few simple economic models, including a simple mathematical framework to capture *technical* abatement that we return to later in the course. While the course will deal further with the concept of abatement costs, we leave the exploration of climate damages and more advanced climate-economic models for other courses.

1 Climate Damages and Abatement Costs

Broadly speaking, *abatement costs* largely refer to the costs associated with abating/reducing emissions. Several strands of literature deals with the assessment of abatement costs, including areas related to natural sciences, environmental engineering, and economics. Given the many different ways there are to avoid emissions altogether, this is quite natural: Emissions can be abated through development of new technologies, but it can also involve adopting new behavior and consumption patterns. Throughout the course, we will look into the most promising and cost-effective ways of delivering emission reductions including renewable electricity, direct/indirect electrification, improving energy efficiency, and more.

To formalize the concept of emission abatement and its related costs, we consider a highly stylized model: Let $F(E)$ denote a production technology that only relies on fossil energy (E).¹ Let p_e denote the cost of extracting one unit of E and ϕ the emissions intensity. Absent any regulation or climate externalities, we would usually assume that a competitive market delivers the outcome that maximizes consumption from:

$$\max C = F(E) - p_e E.$$

Let C^0, E^0 denote the outcomes in this case. The corresponding level of emissions absent abatement is then simply given by

$$M^0 = \phi E^0.$$

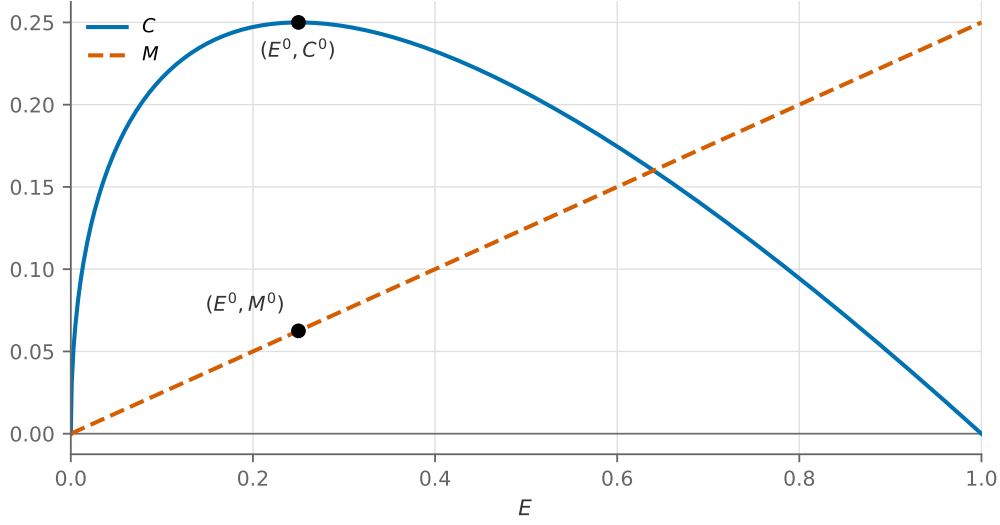
Before we get to the definition of abatement costs, figure 1.1 illustrates our simple model. For small values of E , there is a net gain from increasing the use of fossil energy, while for large values the price of extraction p_e is too great. Thus, absent regulation and climate damages, the optimum is reached at E_0, C_0 as indicated. We refer to this outcome as our *baseline*. Emissions (the dashed line) increase linearly at the rate ϕ for each unit of E .

Given this baseline, we define abatement (A) as reductions in emissions compared to M^0 and the abatement costs (AC) as the reduction in C compared to C^0 :

$$A \equiv M^0 - M$$

¹The following sections gradually build a more complex structure.

Figure 1.1: Baseline scenario, absent damages and regulation



$$AC \equiv C^0 - C.$$

The marginal abatement costs (MAC) are defined as the change in AC induced by a marginal change in A , i.e. the marginal cost of abating one unit of emissions. In our case, this is given by:

$$MAC \equiv \frac{\partial AC}{\partial A} = -\frac{\partial C}{\partial A}.$$

In this small, stylized model everything comes down to how much fossil energy is ultimately used. Thus, the marginal abatement costs follows from

$$MAC = -\frac{\partial C / \partial E}{\partial A / \partial E} = -\frac{F'(E) - p_e}{-\phi} = \frac{F'(E) - p_e}{\phi}. \quad (1)$$

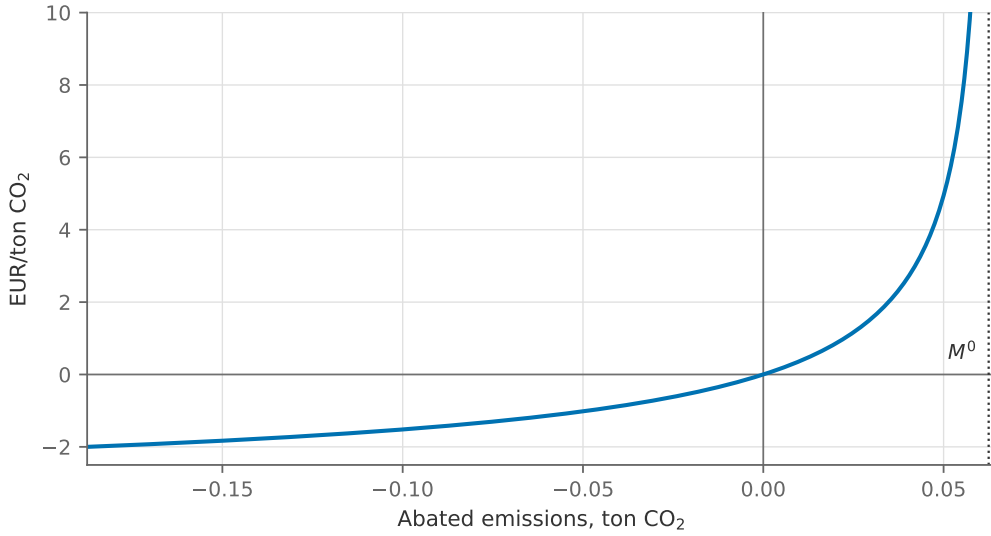
The numerator indicates the *decrease* in potential consumption that comes from decreasing E marginally; the denominator states that lowering E marginally results in ϕ less emissions.

Figure 1.2 illustrates how the MAC function looks with the parameters used in figure 1.1. The figure shows that if CO_2 emissions are not regulated at all ($MAC = 0$), abatement of emissions is per construction in zero. Marginally increasing abatement around this level has very low costs: If we lower the use of E marginally from the baseline (E^0, C^0) , we lose very little consumption (cf. figure 1.1). However, as abatement increases, we have to eliminate consumption of E that is highly productive. Thus, the marginal abatement cost of foregone consumption increases.

In this model, the abatement of emissions comes entirely from consumption/output reduction. However, as we will explore in the following sections, there are other ways of abating emissions. Generally, we distinguish between four abatement channels:²

²You may encounter other ways of explaining the different channels of abatement. For instance, in models without general equilibrium effects and multiple consumption goods, the "sectoral reallocation" channel can also be represented as a form of "input substitution". Also, you may encounter other distinctions between "technical abatement" and "input substitution". For instance, installing a specific technology that delivers better energy

Figure 1.2: Marginal abatement costs



- i. *Output reduction.*
- ii. *Input substitution:* Given the scale of output, a firm can switch to less emission-intensive inputs, e.g. by switching from fossil energy to capital or clean, renewable energy.
- iii. *Technical abatement:* Given inputs and scale of output, some technologies may directly reduce emissions. Carbon Capture and Storage (CCS) and Direct Air Capture (DAC) are good examples of such technologies.
- iv. *Sectoral reallocation:* In richer models with general equilibrium effects and more than one type of consumption good emission reductions may occur if less polluting sectors grow relative to polluting ones.

In the following sections, we will introduce input substitution and technical abatement.

1.1 Climate damages, emissions budgets, and optimal policy

Before we turn to some extensions to the simple model, we will very briefly discuss what constitutes an optimal policy in a scenario with climate damages. This is a gentle introduction to the topic, but the note is by no means comprehensive. Other courses like *Advanced Economics of the Environment and Climate Change* deals with this in a lot more detail.

In climate economics, we refer to the negative externality associated with one ton of CO₂ emissions as the *social cost of carbon*. The discussion of the "correct" social cost of carbon is tremendously complicated and ongoing (see e.g. Drupp et al. 2023). In the current note, we simplify things a lot and simply assume that climate damages relate to the current stock of greenhouse gas (GHG) emissions in the atmosphere. Specifically, we simply assume that

$$\tilde{C} = C - D(M) = F(E) - p_e E - D(M), \quad M = \phi E, \quad (2)$$

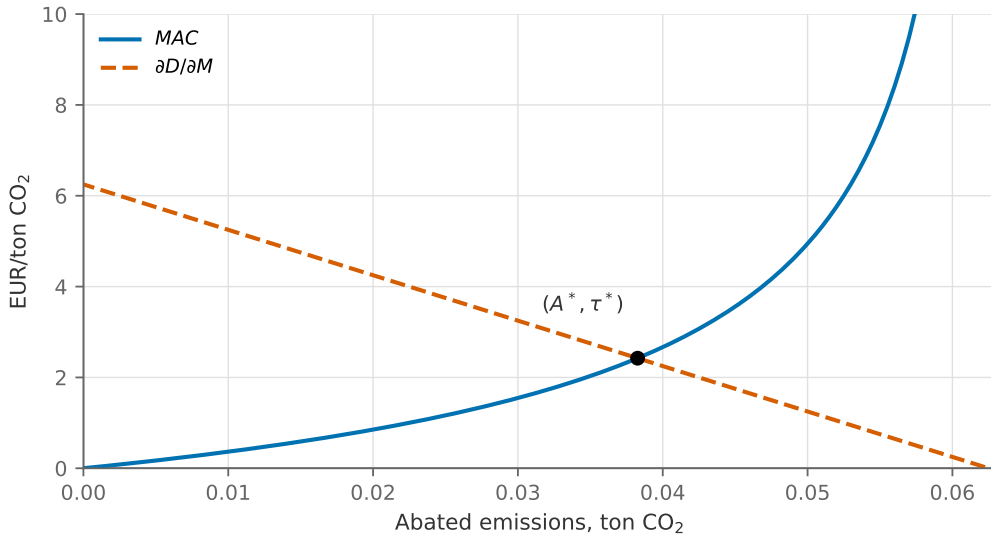
efficiency can be seen as technical abatement or as substitution of capital for energy.

where D measures damages in foregone output. Maximizing $\tilde{C}$ in this case leads to the first order condition

$$\underbrace{\frac{F'(E) - p_e}{\phi}}_{\equiv MAC} = D'(M), \quad (3)$$

which states that *the marginal cost of abating emissions equals the marginal damages from emissions*. Figure 1.3 illustrates the case with increasing marginal abatement costs and decreasing marginal damages (in abated emissions). The optimal level of abatement A^* solves (3) and is associated with a level of marginal abatement costs τ^* that equals marginal damages.

Figure 1.3: Marginal costs and damages



When asked about how to cost effectively mitigate climate change, the first suggestion from most economists is still – arguably – to impose a tax on CO₂ emissions τ^* at the level of the marginal damages (or more broadly, the social cost of carbon). However, there are many reasons why defining and implementing optimal regulation is quite complicated. Amongst the plethora of reasons why, uncertainty of damage estimates and political constraints are right at the top: We are inherently unsure how large damages may be beyond unknown tipping points or planetary boundaries³, and there is no one government body that has authority to align incentives for all to agree on this optimal regulation.

For these reasons – and many more – regulation is often much more complicated than simply imposing a fixed price of CO₂ emissions. Climate targets are often negotiated as national emission budgets that are pursued with a mishmash of technology mandates, CO₂ taxes, subsidies etc..

Throughout the course, we will thus often take the emission target/budget as given and instead focus on the costs associated with reaching this. Instead of the optimization in (2), we will replace

³See e.g. the planetary boundary framework Steffen et al. (2015).

the damage function with an emission target akin to

$$\max C = F(E) - p_e E, \quad M = \phi E, \quad M \leq \overline{M}, \quad (4)$$

where $\overline{M}$ is the target for emissions.⁴ In the following extensions, we will explore both a version with explicit climate damages and one with emission budgets.

2 Adding Technical Abatement

In this first extension, we add specific abatement technologies to the framework in section 1. Specifically, we assume that there are N available technologies that the natural sciences and/or environmental engineers have explained to us. For each of i technology, we assume that we know the following:

- The technology "produces" emission reductions/abatement A_i .
- It has a limit on how much it can abate. This potential is $\theta_i\%$ of all emissions. We use the short-hand $a_i \equiv A_i/(\theta_i M) \in [0, 1]$ to denote the degree of utilization of the technology.
- The costs of using a technology is defined as $AC_i \equiv \theta_i M f_i(a_i)$, where f_i is some convex cost function.⁵

For simplicity, we will in the following assume that $0 < \sum_i \theta_i < 1$.⁶ Adding this to the model from section 1, we now have the following:

$$\begin{aligned} C &= F(E) - p_e E - \sum_i AC_i \\ \tilde{C} &= C - D(M) \\ M &= \phi E - \sum_i A_i \\ A_i &= \phi E \theta_i a_i \\ AC_i &= \phi E \theta_i f_i(a_i). \end{aligned}$$

We can combine these into a single equation in variables E and a_i :

$$\tilde{C} = F(E) - p_e E - D\left(\overbrace{\phi E - \sum_i \underbrace{\phi E \theta_i a_i}_{\equiv A_i}}^{\equiv M}\right) - \sum_i \phi E \theta_i f_i(a_i). \quad (5)$$

⁴Luckily, absent uncertainty about the marginal abatement costs and marginal damages, choosing the correct $\overline{M}$ is equivalent to applying the tax τ^* . This is another field of interesting research that you may want to dive into – e.g. starting with the seminal paper by Weitzman (1974).

⁵A technical note: This way of writing the abatement costs ensures that the *average* abatement costs is always determined by $a_i \in [0, 1]$, which makes the problem scale-invariant. This is a really nice mathematical feature that we will exploit here, but just be aware that this is not always an appropriate assumption.

⁶Something pretty interesting actually happens if $\sum_i \theta_i > 1$: We start being able to extract more CO₂ from the atmosphere *the more* fossil energy consumption that we have. This may sound a bit odd, but it is less crazy than it sounds; this is essentially the case if negative emission technologies rely on "point sources" of CO₂ emissions.

The optimal utilization of technology i is then characterized by the first order condition:

$$\frac{\partial \tilde{C}}{\partial a_i} = D'(M)\phi E\theta_i - \phi E\theta_i f'_i(a_i) := 0,$$

or simply

$$f'_i(a_i) = D'(M). \quad (6)$$

This states that, in optimum, the marginal cost of using technology i should equal the marginal damages from emissions. In addition to this, the optimal level of fossil energy consumption is characterized by:

$$\frac{\partial \tilde{C}}{\partial E} = F'(E) - p_e - \phi D'(M) \left(1 - \sum_i \theta_i a_i \right) - \phi \sum_i \theta_i f_i(a_i) := 0,$$

or, slightly rewritten, as

$$\frac{F'(E) - p_e}{\phi} = D'(M) \underbrace{\left(1 - \sum_i \theta_i a_i \right)}_{(\partial M / \partial E) / \phi} + \sum_i \theta_i f_i(a_i). \quad (7)$$

The left-hand side measures the marginal abatement costs using the *output reduction* channel absent any technical abatement; this was the only available method in the previous section (cf. equation 3). The right-hand side is now a measure of the marginal costs of emissions from output: The first term states that there are damages, but only from the share of emissions that are not abated ($1 - \sum_i \theta_i a_i$). The second term captures the cost of undertaking this additional technical abatement.

As figure 2.1 illustrates, this type of technical abatement generally lowers the marginal cost of consuming fossil energy. The blue line shows the right-hand side of (7) with varying levels of $D'(M)$ *without* any technological abatement (i.e. a diagonal line). The dashed, green line indicates the lower marginal cost of emissions when technology can alleviate some of the problem.

Next, let us turn our attention to the marginal abatement costs. As we did in section 1, we define this as the reduction in consumption from lowering emissions marginally. With the technical abatement, this now becomes:

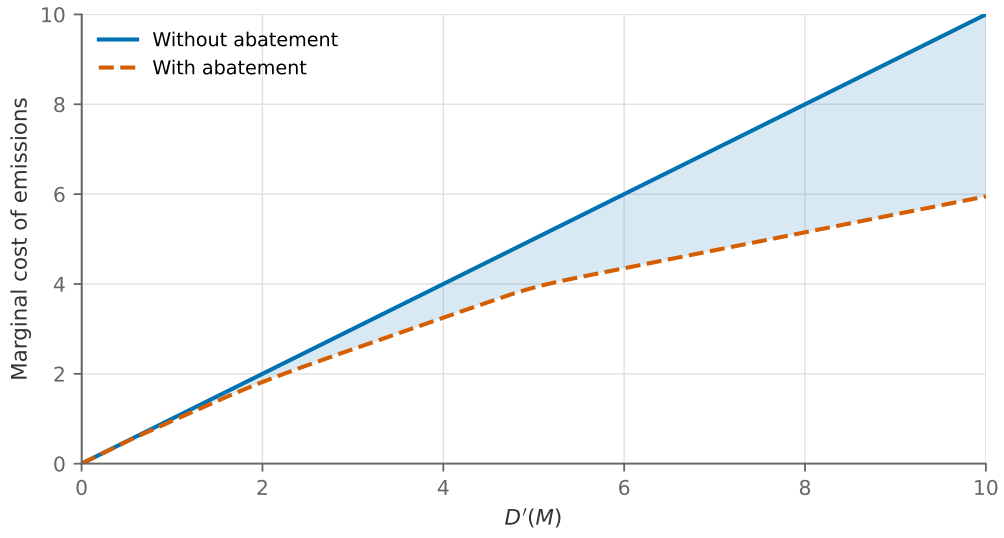
$$MAC = \frac{\partial C}{\partial M} = \frac{\partial C}{\partial E} \frac{1}{\partial M / \partial E}.$$

Using definitions of C , M , and AC_i , we can write this as:

$$MAC = \frac{F'(E) - p_e}{\phi(1 - \sum_i \theta_i a_i)} - \frac{\sum_i \theta_i f_i(a_i)}{1 - \sum_i \theta_i a_i} \quad (8)$$

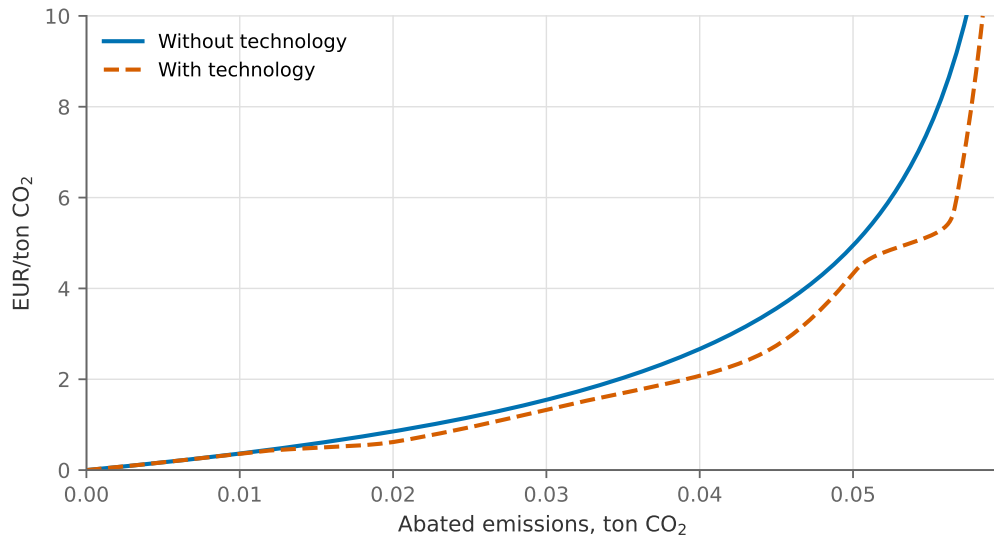
The first term states the same marginal abatement cost as in the simple model, but now with a

Figure 2.1: Technical Effect on Marginal Costs of Emissions



new emission intensity after abatement $\phi(1 - \sum_i \theta_i a_i)$. The second term is a direct technical effect that lowers the marginal abatement costs. Figure 2.2 shows how including abatement technologies generally lowers the marginal cost of abatement.

Figure 2.2: Marginal Abatement Costs with Technical Abatement



References

- Drupp, Moritz A., Frikk Nesje, and Robert C. Schmidt (2023). “Pricing Carbon: Evidence from Expert Recommendations”. In: *American Economic Journal: Economic Policy*.
- Steffen, Will et al. (Jan. 2015). “Planetary Boundaries: Guiding Human Development on a Changing Planet”. In: *Science*.
- Weitzman, Martin L. (1974). “Prices vs. Quantities”. In: *The Review of Economic Studies* October, 41.4, pp. 477–491.

Appendices

A Log-normally distributed costs

As a special case, assume that the abatement technology costs are log-normally distributed with some known average costs, c_i and standard deviation σ_i . In particular, we assume that, in optimum

$$a_i^* = \Phi \left(\frac{\ln(\lambda_i) + \sigma_i^2/2}{\sigma_i} \right), \quad \lambda_i \equiv D'(M)/c_i, \quad (9)$$

where Φ is the standard normal cumulative distribution. This is a log-normal distribution with mean $\mu = -\sigma_i^2/2$. In the following, use $a_i^* \equiv G_i(\lambda_i)$ as a short-hand for this.

Note that this is only a conjecture about the solution to the problem; to fully characterize the solution, though, we need to verify *what* abatement cost function f_i leads to this abatement choice.⁷ To this, we start by noting that the function G_i is related to the abatement cost function through the first order condition (here just repeated from the main text):

$$\frac{\partial f_i}{\partial a_i} = D'(M).$$

If we use the notation $f_i(a_i) = c_i H_i(a_i)$ where H_i is some function to be determined, this means that we can write the first order condition as:

$$H_i'(a_i) = \lambda_i$$

If (9) is indeed the solution (and H_i' is invertible) this means that the functions G_i and H_i are related through:

$$a_i = G_i(\lambda_i) = (H_i'(a_i))^{-1}(\lambda_i).$$

Using this, we can now write the cost function H_i by integrating over the marginal costs and – using the relation to G_i from above – write this as:

$$H_i(a_i) = \int_0^{a_i} H_i'(x) dx = \int_0^{a_i} (G_i)^{-1}(x) dx.$$

Using the change in variables $x = G_i(z)$ and the conjectured solution $a_i = G_i(\lambda_i)$, we can write this as

$$H_i = \int_0^{a_i} (G_i)^{-1}(x) dx = \int_0^{\lambda_i} z \frac{\partial x}{\partial z} dz = \int_0^{\lambda_i} z dG_i(z). \quad (10)$$

A nice feature of this formula is that we can relate this to a distribution's *truncated expectation*.

⁷Technically, we would also need to verify that the solution then exists and that the first order conditions are necessary and sufficient for the solution conjecture to hold. This is pretty straightforward, albeit a bit tedious.

In particular, one can show that:

$$H_i = \int_0^{\lambda_i} z dG_i(z) = a_i \mathbb{E}_{G_i} [z | Z < \lambda_i],$$

where this final part is the truncated/conditional expectation. Assuming that G_i is a log-normal distribution with mean $\mu_i = -\sigma_i^2/2$ the conditional expectation is simply given by (you can look this up online, e.g. on Wikipedia):

$$\begin{aligned} H_i &= a_i \exp\left(\mu_i + \frac{1}{2}\sigma_i^2\right) \frac{\Phi\left(\frac{\ln(\lambda_i) - \mu_i - \sigma^2}{\sigma_i}\right)}{\Phi\left(\frac{\ln(\lambda_i) - \mu_i}{\sigma_i}\right)} \\ &= \Phi\left(\frac{\ln(\lambda_i) - \sigma_i^2/2}{\sigma_i}\right). \end{aligned} \tag{11}$$

This shows that when costs are log-normally distributed, the optimal abatement rate a_i and related costs are given by:

$$\begin{aligned} a_i &= \Phi\left(\frac{\ln(\lambda_i) + \sigma_i^2/2}{\sigma_i}\right) \\ f_i(a_i) &= c_i \Phi\left(\frac{\ln(\lambda_i) - \sigma_i^2/2}{\sigma_i}\right). \end{aligned}$$

You may also verify that for $a_i \rightarrow 1$, the average abatement cost converges to the parameter c_i (look at the unconditional expectation for the log-normal distribution above).